1. A P value is the probability that

A. The null hypothesis is false

B. The null hypothesis is true

C. The observed statistic would have been observed if the null hypothesis were true

D. A statistic at least as extreme as the observed statistic would have been observed if the null hypothesis were true (*answer*)

2. The *most basic* ("bedrock") principle of frequentist inference is:

A. If the null hypothesis were true, we should reject it $\leq \alpha$ of the time (*answer*)

B. We should make "magnitude" errors $\leq \alpha$ of the time

C. We should make "size" errors $\leq \alpha$ of the time

D. We should not compare extremes in two different directions

3. Which of the following would be expected to *increase* the P value in an analysis?

A. A more appropriate model

B. A smaller effect (*answer*)

C. More data

D. All of the above

4. Statistical "significance" is a statement about the study _____ and primarily reflects the _____ of the effects seen.

A. data; clarity (*answer*)

B. data; importance

C. system; clarity

D. system; importance

5. `c("Chocolate", "Vanilla", "Strawberry")` is an example of _____ data

A. binary

B. interval

C. ordinal

D. nominal (*answer*)

E. ratio

6. In general, interval data is best viewed on a _____ scale and ratio data is best viewed on a _____ scale

A. linear; log

B. log; linear

C. linear; linear or log (*answer*)

D. linear or log; linear

Alternative option A gets 0.8 points

7. Tidyverse functions generally take a data table _____ because this is more convenient for _____

A. as the first argument; piping (*answer*)

B. as the first argument; selecting variables

C. from a variable called `dat`; piping

D. from a variable called `dat`; selecting variables

8. Which of the following does *not* always need to be specified in order to produce a ggplot showing data?

A. aesthetic mapping

B. data

C. geom

D. scale (*answer*)

9. Compared to classical statistical tests, the permutation test discussed in class requires _____ assumptions and is often _____ powerful.

A. more; more

B. more; less

C. fewer; more

D. fewer; less (*answer*)

Alternative option C gets 0.8 points

10. "Asymptotically valid" means that a test will satisfy the paradigm for error rate for _____ sample size

A. any

B. a large enough (*answer*)

C. a small enough

D. a not-too-large and not-too-small

```
big <- dat |> filter(weight > 185)
bigger <- dat |> filter_out(weight <= 185)
```

Use the code above for the next three questions

11. Each line uses _____ as the _____ for a filtering function

A. **`dat`; first argument** (*answer*)

B. `weight`; first argument

C. `dat`; output

D. `weight`; output

12. *Choose the most precise correct answer.* `big` and `bigger` will be the same:

A. As long as `weight` can exceed 185

B. If `dat` has no NAs

C. If the `weight` variable has no NAs (*answer*)

D. Virtually always

E. Virtually never

13. `big` will be a _____ with the same _____

A. table; columns as `dat` (*answer*)

B. table; columns as `weight`

C. vector; length as `dat`

D. vector; length as `weight`

14. A data set reflects temperatures of 8, 7, NA, 9 °C. You probe the data set using `summary()` and then construct a data set of means using `summarise(temp=mean(temp))`. You would expect `summary` to show a mean of _____ and `summarise` to show a mean of _____.

A. 6; 6

B. 6; NA

C. NA; 6

D. 8; NA (*answer*)

E. NA; 8

15. Which of the following is *not* a key principle according to this course? Your code should be:

A. clear

B. efficient (*answer*)

C. replicable

D. shareable

16. The code `dat |> select(name, macid)` selects

A. rows named `name` or `macid`

B. columns named `name` or `macid` (*answer*)

C. rows that have non-missing information about `name` or `macid`

D. columns that have non-missing information about `name` or `macid`

```
meds <- (dat
    |> summarise(phos=median(phos), nit=median(nit)
        , .by=species
    )
)
```

Use the code above for the next two questions

17. In the code above, the `.by=species` argument tells R to

A. Exclude the variable `species` from the results

B. Find the median species for each of `phos` and `nit`

C. Include observations where the variable `species` is `NA`

D. Report a median for each species (*answer*)

18. *If the `.by` argument was dropped,* this code would produce a table with:

A. The same number of rows and columns as dat

B. The same number of rows as dat and two columns

C. One row and the same number of columns as dat

D. One row and two columns (*answer*)

```
aMean <- dat |> filter(is.na(height)) |> summarise(height=mean(height))
bMean <- dat |> filter(!is.na(height)) |> summarise(height=mean(height))
cMean <- dat |> summarise(height=mean(height))
```

Use the code above for the next two questions

19. In general:

A. `aMean` is more useful than `bMean`

B. **`bMean` is more useful than `aMean`** (*answer*)

C. `aMean` and `bMean` are useful in different contexts

D. `aMean` and `bMean` are not useful

20. Not filtering NAs (as in `cMean`) is _____ if you always want an answer (e.g., to get a general overview of `height`), and _____ if you want to avoid ignoring missing data points.

A. good; good

B. good; not good

C. not good; good (*answer*)

D. not good; not good

```
> ?runif
...
runif(n, min = 0, max = 1)
```

21. According to the R help line above, the second argument of `runif` is _____, and it is _____

A. min; required

B. min; not required (*answer*)

C. max; required

D. max; not required

```
dat <- (dat
    |> ______(
        totalArea = Land + Water
        , density = Population/totalArea
        , check = totalArea-Total
    )
)
```

Use the code above (from the provinces code discussed in class) for the next two questions

22. Which verb does the blank in the code represent?

A. count

B. filter

C. mutate (*answer*)

D. select

E. summarise

23. The purpose of `check` is

A. To make sure that the three area columns in the original table are consistent (*answer*)

B. To rank the provinces by density

C. To see which provinces have more land than water

D. To see which provinces have more area than people

24. If you find a mistake by looking for outliers, you should

A. improve your upstream quality control (*answer*)

B. remove it and move on

C. test your conclusions with and without the point

D. transform your data

25. You should not remove an outlier from an analysis of your data unless it:

A. is flagged as an error using a documented quality-control pathway (*answer*)

B. passes a pre-specified statistical test for being an outlier

C. meets both of these criteria

D. you should not remove any data from any analysis